Final Report

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INDIVIDUAL CONTRIBUTIONS :

Member Name	Core Role	Descriptive Contribution
Ornima Hafizur	Admin & Authentication	Implemented the administrative access path and authentication-related workflow. Connected authenticated sessions with role-specific dashboards and built administrative controls over users, bookings, drivers, vehicles, and system reports.
Md. Shihabur Rahman Shihab	Driver Module	Developed assigned-booking, profile management, availability tracking, earnings, and performance analytics to give drivers a dedicated operational view.
Abdul Ahad	Vehicle Owner Module	Implemented fleet management operations, including vehicle CRUD actions, booking management, driver assignments, earnings tracking, and ratings.
Anisha Zaman	Customer Module	Developed the customer rental workflow covering vehicle discovery, booking, payment processing, booking history, reviews, comparison, and profile management.

ABSTRACT

RideRent Pro is a comprehensive, role-based vehicle rental management platform engineered to synchronize operational workflows among four primary platform actors: customers, vehicle owners, drivers, and platform administrators. Developed using PHP 8.0+ and MySQL, the system documents end-to-end authentication mechanisms, fleet lifecycle management, real-time availability tracking, transaction verification, driver dispatching, multi-criteria search filtering, interactive dashboards, and business intelligence analytics. This technical report details the software architecture, entity-relationship models, operational logic, security considerations, testing methodologies, current limitations, and an extensive future implementation roadmap.

INTRODUCTION & PROJECT BACKGROUND :

Modern vehicle rental services operate within dynamic environments requiring multi-party coordination, instant availability verification, and strict transaction tracking. In traditional manual or decentralized operational setups, rental businesses suffer from scheduling conflicts, duplicated reservations, unverified payments, and poor inventory visibility. RideRent Pro addresses these systemic challenges by offering a centralized, multi-tenant digital hub where all operational state changes are synchronized through a unified database architecture.

The system architecture cleanly separates configuration, global utility functions, security guards, shared components, and dedicated role modules. Each role is granted a tailored user interface designed around specific operational permissions and tasks:

- **Customer Module:** Provides an intuitive storefront enabling users to filter vehicles, check date availability, select drivers, choose payment modes (digital vs. manual), view active booking histories, submit reviews, and compare vehicle metrics.
- **Owner Module:** Equips vehicle fleet owners with full lifecycle inventory management (CRUD), reservation tracking, revenue and earnings performance analytics, and local driver allocation capabilities.
- **Driver Module:** Delivers an operational dispatch portal where drivers manage incoming assignment requests, toggle availability status, track trip earnings, and view personal customer feedback metrics.
- **Administrator Module:** Grants platform supervisors centralized governance over user credentials, fleet approvals, driver verification, dispute resolution, financial auditing, and system-wide reporting analytics.

OBJECTIVES & SYSTEM REQUIREMENTS :

Primary Objectives

1. **Centralization:** Eliminate operational silos by unifying customer, owner, driver, and admin operations into a single scalable database ecosystem.
2. **Automation:** Streamline rental workflows by automating availability checking, pricing logic, transaction tracking, and driver assignment routing.
3. **Accountability:** Establish clear audit trails using transaction reference codes, payment states, transparent rating systems, and activity logging.

4. **Scalability:** Build a foundation capable of expanding into advanced smart services, including mobile applications, automated verification, and AI-driven optimizations.

Technology Stack & Specifications

- **Backend Engine:** PHP 8.0+ executing modular business logic handlers and controller scripts.
- **Database Engine:** MySQL 5.7+ utilizing relational schemas, foreign key constraints, and transactional storage engines.
- **Frontend Layer:** Responsive HTML5, custom CSS styling, JavaScript for client-side dynamic DOM interactions, and Font Awesome icon suites.
- **Server Infrastructure:** Compatible with Apache or Nginx server environments supporting modern PHP session security features.

SYSTEM ARCHITECTURE & DATA MODEL :

Layered System Architecture

RideRent Pro implements a structured three-tier web application architecture designed to maximize maintainability, security, and parallel modular development:

- **Presentation Layer (UI/UX):** Responsive, role-aware client templates rendered dynamically via PHP and structured HTML/CSS. Interfaces strictly adjust controls based on the logged-in session context.
- **Application / Business Logic Layer:** Controller scripts handle requests, enforce server-side authorization checks, process session routines, execute business rules, and calculate rental pricing formulas.
- **Persistence / Data Layer:** A relational MySQL database storing user profiles, vehicle metadata, active bookings, payment transactions, and feedback records with enforced referential integrity.

Core Relational Schema & Entities

The data model comprises seven core relational entities structured to handle complex multi-entity relationships:

- `admin`: Stores administrative profile data and system privilege flags.
- `customer`: Maintains customer contact details, profile preferences, and account history.
- `vehicle_owner`: Links fleet owners with business verification profiles and financial summaries.
- `driver`: Records driver licensing info, active availability states, and accumulated rating scores.
- `vehicle`: Contains vehicle specifications, category classifications, daily rental rates, and approval status flags.
- `booking`: Serves as the central transaction junction linking customer, vehicle, driver, rental date ranges, pricing, and payment states.
- `reviews`: Stores rating values (1–5 stars) and detailed textual feedback tied directly to completed bookings.

Operational Logic & Mathematical Rules

The application enforces continuous mathematical logic to preserve system integrity across reservations and financial accounting.

1. Availability Checking Formula

To prevent double-booking, a requested rental interval $[s, e)$ for a specific vehicle is validated against all existing confirmed bookings i having intervals $[s_i, e_i)$. The vehicle is strictly available if and only if:

$$\forall i,\ (e \le s_i \ \&lor;\ s \ge e_i)$$

If any overlap occurs, the reservation attempt is blocked dynamically at the controller layer.

2. Total Rental Cost Model

The pricing engine calculates total customer billing using vehicle rates, optional driver fees, and late return penalties:

$$C_{base} = R_{vehicle} \times N_{days}$$
$$C_{driver} = R_{driver} \times N_{days}$$
$$C_{late} = max(0, N_{late}) \times R_{late}$$

IMPLEMENTATION & CORE WORKFLOWS :

Authentication & Session Routing

Authentication forms the entry point of the application. Upon visiting the landing controller (`index.php`), the server initializes a secure session via `session_start()` and inspects session state keys:

- `$_SESSION['admin_id']` → Routes to `/admin/dashboard.php`
- `$_SESSION['owner_id']` → Routes to `/owner/dashboard.php`
- `$_SESSION['driver_id']` → Routes to `/driver/dashboard.php`
- `$_SESSION['customer_id']` → Routes to `/customer/dashboard.php`

Unauthenticated visitors are presented with public storefront controls, vehicle browsing options, and login/registration triggers.

End-to-End Reservation Workflow

The standard customer reservation workflow follows a structured six-stage linear state transition:

1. **Discovery & Filtering:** Customers search inventory by location, seating capacity, price range, and vehicle type.
2. **Availability Validation:** The system verifies the selected vehicle's calendar against active DB reservations.
3. **Driver Option Selection:** The user chooses self-drive or requests a professional driver from available platform drivers.
4. **Reservation Confirmation:** A pending booking record is instantiated within the database with locked pricing variables.
5. **Payment Processing:** The customer completes payment choosing digital gateway channels (bKash, Nagad, Card) or manual cash pickup.
6. **Receipt & Dispatch:** A formal receipt containing a unique transaction reference ID is rendered, and dispatch notices are routed to the owner and driver.

Payment Tracking States

To accommodate diverse regional transaction practices, RideRent Pro supports a robust payment state machine:

Payment State	Trigger Condition	System Action
Pending	Booking created under "Pay Later" option	Inventory temporarily reserved; driver dispatch paused.
Paid	Digital payment completed / Manual verified	Booking status confirmed; thermal receipt generated.
Refunded	Booking cancelled within allowable grace period	Inventory unlocked; owner earnings adjusted.

TESTING, SECURITY & QUALITY ASSURANCE :

System Test Matrix

To verify structural reliability, business logic correctness, and secure route access, the application was subjected to standardized functional test scenarios.

Testing Area	Test Scenario & Execution	Expected System Outcome

Authentication	Login with valid/invalid credentials; direct URL injection attempt without active session.	Successful redirection to assigned role dashboard or immediate session bounce to login.
Availability	Submitting a booking request overlapping an existing confirmed date range for the same vehicle.	Validation failure triggered; reservation blocked with user conflict notice.
Booking Lifecycle	Creating, updating, paying, and completing full rental journey.	Correct state transitions from Pending to Paid; automated receipt generation.
Driver Dispatch	Assigning driver to active booking while driver is set to "Unavailable".	Assignment rejected; only active and available drivers selectable.
Authorization Guard	Customer account attempting direct HTTP GET request to `/admin/reports.php`.	Server-side authorization check catches privilege mismatch; returns HTTP 403 / redirect.

Security Hardening Parameters

Platform integrity is maintained using industry-standard web security practices aligned with OWASP guidelines:

- **Credential Security:** Plaintext passwords are strictly prohibited. Passwords are hashed using standard `password_hash()` implementation with strong default algorithms.
- **SQL Injection Mitigation:** Database queries execution utilizes parameterized prepared statements for all dynamic inputs, eliminating SQL injection vulnerabilities.
- **Session Hijacking Protection:** Session tokens undergo automatic regeneration post-authentication, preventing session fixation attacks.
- **Cross-Site Request Forgery (CSRF):** Form submissions incorporate anti-CSRF tokens to validate origin authenticity.

CURRENT SCOPE & LIMITATIONS :

While RideRent Pro delivers a feature-complete baseline platform for multi-role vehicle rental coordination, academic and resource boundaries defined specific operational limits during the initial implementation phase.

Identified Technical Limitations

- **Gateway Integration:** The payment processing module simulates transaction verification via digital transaction reference IDs rather than direct live REST API settlement with external financial institution gateways.
- **UI Runtime Artifacts:** Certain layout visualizations represent static web client reconstructions rather than real-time live browser canvas captures.
- **Notification Infrastructure:** System alerts currently rely on in-app dashboard notification badges rather than external automated cellular SMS or SMTP email push services.
- **Location Services:** Vehicle availability filtering uses manual geographic region selection rather than live

real-time GPS positioning metrics.

> **Evaluation Summary:** The current project scope successfully fulfills all core requirements for CSE 299, establishing a scalable architectural foundation ready for enterprise-grade feature expansion.

FUTURE IMPLEMENTATION ROADMAP :

To evolve RideRent Pro into an enterprise-grade commercial vehicle mobility platform, a comprehensive multi-phase future implementation roadmap has been designed. These enhancements focus on smart automation, real-time tracking, hardware integration, and advanced business intelligence.

1. AI-Powered Vehicle Recommendation Engine

Future versions will incorporate a Machine Learning (ML) recommendation model that analyzes historical customer booking patterns, budget preferences, seasonal trends, and trip types (e.g., family vacation, business commute, off-road) to suggest optimal vehicle choices dynamically.

$$Score(v) = \sum (w_k \times f_k(v))$$

Where $f_k(v)$ evaluates similarity between customer profile vector metrics and vehicle feature attributes v.

2. Contactless QR-Code Verification System

To streamline vehicle pickup and return workflows, an automated QR-code verification system will be implemented:

- Upon booking confirmation, a unique encrypted QR token is generated and stored on the customer's dashboard.
- Vehicle owners or assigned drivers scan the customer's QR code using the mobile portal upon physical handover.
- The system instantly verifies the transaction token, validates identity proof, records time-stamped digital handovers, and unlocks the vehicle state automatically.

3. Real-Time Telematics & GPS Fleet Tracking

Integrating IoT-enabled GPS OBD-II telematics hardware into registered vehicles will unlock live operational capabilities:

- **Live Route Mapping:** Customers and vehicle owners can track real-time trip progress on interactive vector maps.
- **Geo-Fencing Alerts:** Automatic system notifications triggered if a vehicle crosses designated operational boundaries or restricted zones.
- **Remote Diagnostics:** Continuous monitoring of engine health metrics, fuel levels, tire pressure, and

driving behavior analytics (sudden braking, overspeeding).

4. Mobile Application & Push Notification Suite

Expanding platform accessibility beyond web browsers through dedicated cross-platform mobile applications (iOS / Android) built using React Native or Flutter:

- Instant SMS & Email push notifications for trip dispatch updates, payment receipts, and driver arrival alerts.
- In-app real-time messaging chat channel allowing secure communication between customers, assigned drivers, and owners without disclosing private mobile numbers.

5. Automated Maintenance & Branch Management

Enterprise multi-branch support enabling commercial rental agencies to manage distributed physical hubs efficiently:

- **Branch Allocation:** Vehicle drop-off at different physical branch locations with automated inter-branch transfer tracking.
- **Predictive Maintenance Scheduling:** Automated scheduling of routine vehicle oil changes, tire rotations, and inspections based on accumulated odometer mileage tracked during rentals.

Summary Matrix of Future Modules

Future Feature	Target Domain	Core Expected Benefit
AI Recommendations	Smart Services	Increases booking conversion rates through personalized vehicle suggestions.
QR Handover Validation	Operations & Security	Eliminates physical paperwork and prevents unauthorized vehicle release.
IoT GPS Telematics	Fleet Management	Enhances asset security via live tracking and boundary breach alerts.
Multi-Branch Logistics	Enterprise Expansion	Enables seamless point-to-point one-way rental models across different cities.
Predictive Maintenance	Fleet Quality Assurance	Reduces breakdown risks by auto-scheduling service based on mileage logs.

CONCLUSION :

RideRent Pro presents a robust, academically grounded, and architecturally coherent digital solution for modern vehicle rental management. By clearly separating platform administrators, vehicle fleet owners, drivers, and customers, the platform resolves systemic challenges associated with booking conflicts, payment ambiguity, and poor inventory visibility. The underlying modular PHP and MySQL architecture ensures high maintainability, secure session handling, and transparent transaction auditing across all operational touchpoints.

Furthermore, the detailed roadmap outlining future integrations—such as AI-driven recommendations, contactless QR verification, IoT telematics, mobile application suites, and predictive maintenance—positions RideRent Pro as a scalable foundation that can evolve into a full-scale commercial transport management ecosystem.

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